Appendix

Flexible regulation as a double-edged sword: The case of the organic seed standard

Survey sampling method

Surveys were collected using a mix of convenience and random sampling, with the approach varying across the three time periods.

- 2010: In the 2010 survey period, data were collected by convenience sample only, shared by 23 organizations, including non-profits and foundations, organic certifiers, and departments of agriculture. Surveys were distributed [**JARED please let me know, mail/email process**]. We received a total of 1,029 responses, 545 of which included complete enough data for our analysis.
- 2015: In the 2015 survey period, data were collected by both random and convenience sampling. For the random sample, 800 certified organic producers were randomly selected from the approximately 12,000 organic producers that use seed recorded in the NOP database. To encourage responsiveness, three follow-up emails were sent following an initial email, paper copies were mailed on request, and individual phone calls were made to follow-up. From these 800, 197 (25%) responded and 127 (16%) were complete enough to use in our analyses. For the convenience sample, surveys were shared by 41 organizations, again including non-profits and foundations, organic certifiers, and departments of agriculture. Surveys were distributed [**JARED please let me know, mail/email process**]. We received a total of 1,148 responses from the convenience sample, 672 of which included complete enough data for our analysis. In total we had 799 usable responses.
- 2020: In the 2020 survey period, again data were collected by both random and convenience sampling. For the random sample, 2,000 certified organic producers were randomly selected from the the NOP database. To encourage responsiveness, the survey was offered as both an online version and a paper questionnaire, and respondents had the option of completing either. After an initial postcard invitation, non-respondents received follow-up contacts by mail, email, and phone. Survey distribution was managed by Washington State University's Social and Economic Sciences Research Center. From these 2,000 surveys, 448 (22%) responded (206 web surveys and 242 paper surveys) and 193 (10%) were complete enough to use in our analyses. For the convenience sample, surveys were distributed to all remaining certified organic operations over email. For the convenience sample, surveys were shared by OFRF and OSA, social and traditional media, and through organic certifiers and organic farming membership organizations **UPDATE**. We received a total of 611 responses from the convenience sample, 215 of which included complete enough data for our analysis. In total we had 408 usable responses.

The breakdown between random and convenience sampling used in our analysis is summarized in Table 1 and Table 2. We also consider the effect of sampling method, as well as survey distributor, through models in Table 4.

Table 1: Convenience and random samples

Year	Total complete responses	Convenience responses (%)	Random responses (%)
2010	545	545 (100 %)	0 (0 %)
2015	799	672 (84 %)	127 (16 %)
2020	408	215 (53 %)	193 (47 %)

Table 2: Number and percentage of region-crop combinations within convenience and random samples in 2015 and 2020 survey periods

		2015		2020	
Region	Main crop	Convenience	Random	Convenience	Random
North Central	Field	229 (34%)	20 (16%)	44 (20%)	62 (32%)
North Central	Forage	63 (9%)	4 (3%)	10 (5%)	17 (9%)
North Central	Vegetables	71 (11%)	8 (6%)	25 (12%)	19 (10%)
Northeast	Field	20 (3%)	8 (6%)	11 (5%)	21 (11%)
Northeast	Forage	24 (4%)	8 (6%)	4 (2%)	10 (5%)
Northeast	Vegetables	55 (8%)	24 (19%)	31 (14%)	18 (9%)
South	Field	13 (2%)	1 (1%)	9 (4%)	4 (2%)
South	Forage	7 (1%)	1 (1%)	3 (1%)	NA
South	Vegetables	47 (7%)	6 (5%)	12 (6%)	11 (6%)
West	Field	22 (3%)	9 (7%)	10 (5%)	10 (5%)
West	Forage	21 (3%)	9 (7%)	9 (4%)	5 (3%)
West	Vegetables	100 (15%)	29 (23%)	47 (22%)	16 (8%)

Sample representativeness

While the mix of sampling methods creates a challenge for determining the sample's representativeness of the population, we aim to demonstrate that our sample is representative by comparing the geographic distribution and approximate average farm size of our sample to USDA Agricultural Census data. We use USDA Census data from 2011, 2016, and 2021 to align with our 2010, 2015, and 2020 survey periods.

Across the three survey periods, the number of respondents represented approximately 6.3%, 5.9%, and 2.5% of the total number of organic farmers identified in the USDA Census in the related years, though these timelines are not exactly set to match.

In Table 3 we compare the number of operations in our sample (n) to the population (N) by year and region. For each census period we calculated the standard deviation of the percentage of operations represented, which we used to identify regions that are over or under-represented (mean of % operations represented +/- sd of % operations represented). Over-represented year-region combinations are signaled by green text, while under-represented year-region combinations are signaled by red text. Only in 2011 do we see variation in representation across regions, with the North Central and Western regions over-represented and the Northeast under-represented.

We also calculate the mean acreage in our sample, as well as the mean acreage according to the USDA census (N operations/total acreage) for a given year-region combination. Using the mean here provides limited information, given that farm sizes vary considerably in ways that are not captured in aggregate data available in the USDA census. Though crude, these averages suggest relative similarity in average farm size.

We calculate farm size as acreage planted to 3 major crop categories – probably underestimates compared to USDA, because they may account for all land (look into that?) Can we put categorical farm size into the model to double check largest farms aren't pulling up effect

Table 3: Sample representativeness compared to USDA Censuses (2011, 2016, 2021)

Year	Region	Operations in sample (n)	Operations in USDA census (N)	Percent of operations represented (n/N)	Mean acreage in sample	Mean acreage in USDA Census	Mean acreage difference
2010	North Central	304	2986	10.2	195	212	-17
2010	Northeast	51	1897	2.7	142	142	0
2010	South	51	540	9.4	105	180	-75
2010	West	139	3296	4.2	194	248	-54
2015	North Central	395	4689	8.4	307	184	123
2015	Northeast	139	3246	4.3	142	135	7
2015	South	75	1096	6.8	144	215	-71
2015	West	190	4529	4.2	358	260	98
2020	North Central	177	5824	3	323	204	119
2020	Northeast	95	3888	2.4	205	143	62
2020	South	39	1209	3.2	847	279	568
2020	West	97	5273	1.8	329	284	45

Checking for biases induced by data collection methods

In order to test how our data collection methods may bias the sample, we consider the effect of the sampling approach and survey distributor.

Regarding concerns of sampling approach, we model the effect of sampling method with our 2015 and 2020 survey data (the only survey periods where we had both sampling approaches), including a term for responses that were gathered by random sampling compared to convenience sampling (Table 4, Model 1). We see a small and insignificant effect for sampling method, and negligible changes in the results between the models run with and without the sampling consideration (Table 4, Model 2).

Regarding concerns of survey distributor (i.e. that the distribution by Organic Seed Alliance (OSA) elicited responses from a select type of organic farmer), we model the effect of sampling method with our 2010 and 2015 survey data (the only survey periods that identified specific distributors), including a term for responses that were gathered by OSA compared to all others (Table 4, Model 3). We see a small and insignificant effect for distributor, and negligible changes in the results between the models run with and without the sampling consideration (Table 4, Model 4).

Table 4: Model checks for sampling and distributor effects

	Proportion of organic land planted with conventional seed			
	Including sampling method (2015, 2020)	Without sampling method (2015, 2020)	Including survey distributor (2010, 2015)	Without survey distributor (2010, 2015)
	(1)	(2)	(3)	(4)
Barrier: Organic availability	0.181***	0.181***	0.131***	0.132***
	(0.043)	(0.043)	(0.042)	(0.042)
Barrier: Buyer requirements	0.117***	0.115***	0.122***	0.120***
	(0.041)	(0.041)	(0.038)	(0.038)
Barrier: Undesirable traits	0.112**	0.111**	0.123***	0.124***
	(0.045)	(0.045)	(0.041)	(0.041)
Barrier: Insufficient quantity	0.052	0.054	0.012	0.011
	(0.043)	(0.043)	(0.038)	(0.038)
Year: 2015 (vs. 2010)			−0.191**	−0.187**
			(0.076)	(0.076)
Values organic seed	−0.219***	−0.218***	−0.215***	−0.213***
	(0.042)	(0.042)	(0.037)	(0.037)
Farm size (log acres)	0.094***	0.096***	0.115**	0.113**
	(0.034)	(0.034)	(0.047)	(0.046)
Crop diversity	−0.020	−0.014	−0.041	−0.042
	(0.041)	(0.040)	(0.038)	(0.038)
Seed saved or traded (%)	−0.046	−0.045	−0.089***	−0.087**
	(0.038)	(0.038)	(0.034)	(0.034)
Certifiers request organic seed use	0.019	0.019	0.057	0.056
	(0.038)	(0.038)	(0.036)	(0.036)
Forage crops (vs. Field)	0.991***	0.992***	1.019***	1.021***
	(0.114)	(0.114)	(0.105)	(0.105)
Vegetable crops (vs. Field)	0.314***	0.323***	0.331***	0.339***
	(0.100)	(0.099)	(0.098)	(0.097)
Western region (vs. North Central)	0.234**	0.229**	0.303***	0.332***
	(0.104)	(0.104)	(0.102)	(0.098)
Northeastern region (vs. North Central)	0.285***	0.277***	0.166	0.202*
	(0.106)	(0.105)	(0.117)	(0.112)
Southern region (vs. North Central)	0.032	0.033	0.131	0.150
	(0.137)	(0.137)	(0.131)	(0.130)
Random data collection (versus convenience)	−0.055			
	(0.087)			
Distributor: OSA (vs. Other)			0.083	
			(0.081)	
Constant	−0.916***	−0.931***	−0.763***	−0.752***
	(0.071)	(0.067)	(0.075)	(0.074)
Observations	1,207	1,207	1,344	1,344
R^2	0.252	0.252	0.248	0.247
Log Likelihood	1,731.220	1,731.018	1,658.831	1,658.309

Note: *p<0.1; **p<0.05; ***p<0.01

Certifiers' role

An additional consideration in our analysis is the effect of certifier. We expect certifiers to enforce the regulation differently **(cite)**, but we do not have data that allows us to specifically identify each producers' certifier in our data for use in our main analysis. However, we do have some suggestion of growers' certifier based on our data collection process in certain survey periods. When data are available, we assume that respondents who completed the survey distributed by a certain certifier are certified by them. In Table 5, Model 1 we model the effect of five distinct certifiers and one 'other' category, compared against compared to a baseline of the most popular certifier in our limited dataset. We see a small and insignificant effect for different certifiers compared to a baseline certifier group, and negligible changes in the results between the models run with and without the sampling consideration (Table 3, Model 2).

Table 5: Model checks for certifier effects

	Proportion of organic land planted with conventional seed	
	(1)	(2)
Barrier: Organic availability	0.156**	0.159**
	(0.070)	(0.070)
Barrier: Buyer requirements	0.087	0.087
	(0.062)	(0.061)
Barrier: Undesirable traits	0.042	0.040
	(0.069)	(0.069)
Barrier: Insufficient quantity	0.024	0.023
	(0.064)	(0.064)
Year: 2015 (vs. 2010)	−0.298	−0.345**
	(0.240)	(0.138)
Year: 2020 (vs. 2010)	−0.394	−0.450*
	(0.383)	(0.255)
Values organic seed	−0.164***	−0.163***
	(0.063)	(0.063)
Farm size (log acres)	0.150***	0.153***
	(0.044)	(0.043)
Seed saved or traded (%)	−0.252**	−0.260**
	(0.106)	(0.104)
Certifiers request organic seed use	−0.083	−0.070
	(0.063)	(0.060)
Forage crops (vs. Field)	0.859***	0.843***
	(0.162)	(0.159)
Vegetable crops (vs. Field)	0.678***	0.651***
	(0.206)	(0.201)
Western region (vs. North Central)	0.397	0.362**
	(0.274)	(0.179)
Northeastern region (vs. North Central)	0.384	0.340
	(0.255)	(0.236)
Southern region (vs. North Central)	0.290	0.059
	(0.395)	(0.289)
Certifier Other	−0.068	
	(0.334)	
Certifier 24	−0.346	
	(0.450)	
Certifier 2	−0.067	
	(0.309)	
Certifier 27	−0.002	
	(0.265)	
Certifier 42	0.181	
	(0.247)	
Certifier 7	0.011	
	(0.348)	
Constant	−1.049***	−1.008***
	(0.286)	(0.274)
Observations	487	487
R^2	0.240	0.238
Log Likelihood	625.536	624.823

Note: *p<0.1; **p<0.05; ***p<0.01

Full model results

Table 6: Final model results

	Proportion of organic land planted with conventional seed
Barrier: Organic availability	0.170^{***}
	(0.037)
Barrier: Buyer requirements	0.116^{***}
	(0.033)
Barrier: Undesirable traits	0.106^{***}
	(0.036)
Barrier: Insufficient quantity	0.0001
	(0.034)
Year: 2015 (vs. 2010)	-0.207^{***}
	(0.075)
Year: 2020 (vs. 2010)	-0.363^{***}
	(0.092)
Values organic seed	-0.189^{***}
	(0.033)
Farm size (log acres)	0.096^{***}
	(0.021)
Crop diversity	-0.121^{**}
	(0.054)
Seed saved or traded (%)	-0.095^{***}
	(0.031)
Certifiers request organic seed use	0.028
	(0.032)
Forage crops (vs. Field)	0.919^{***}
	(0.093)
Vegetable crops (vs. Field)	0.487^{***}
	(0.103)
Western region (vs. North Central)	0.363^{***}
	(0.085)
Northeastern region (vs. North Central)	0.285^{***}
	(0.093)
Southern region (vs. North Central)	0.161
	(0.113)
Constant	-0.970^{***}
	(0.144)
Observations	1,752
R^2	0.244
Log Likelihood	2,265.746
Note:	*p<0.1; **p<0.05; ***p<0.01

Model results over time

Table 7: Model results by year

	2010	2015	2020
Barrier: Organic availability	−0.098	−0.178***	−0.278***
	(0.074)	(0.051)	(0.075)
Barrier: Undesirable traits	−0.130**	−0.116**	−0.124
	(0.060)	(0.055)	(0.077)
Barrier: Insufficient quantity	0.039	−0.042	−0.052
	(0.056)	(0.052)	(0.076)
Values organic seed	0.175***	0.212***	0.116
	(0.053)	(0.048)	(0.073)
Farm size (log acres)	−0.101**	−0.122***	−0.102**
	(0.039)	(0.029)	(0.045)
Crop diversity	0.145**	0.019	0.131*
	(0.058)	(0.051)	(0.075)
Seed saved or traded (%)	−0.205*	−0.065	0.048
	(0.115)	(0.092)	(0.135)
Certifiers request organic seed use	0.188***	0.024	0.117
	(0.057)	(0.043)	(0.079)
Forage crops (vs. Field)	−0.803***	−1.233***	−0.514**
	(0.158)	(0.139)	(0.203)
Vegetable crops (vs. Field)	−0.383**	−0.777***	−0.295
	(0.186)	(0.151)	(0.214)
Western region (vs. North Central)	−0.610***	−0.184	−0.345**
	(0.154)	(0.126)	(0.176)
Northeastern region (vs. North Central)	−0.092	−0.255*	−0.300*
	(0.208)	(0.134)	(0.168)
Southern region (vs. North Central)	−0.398**	−0.015	−0.089
	(0.202)	(0.167)	(0.235)
Constant	0.865***	1.173***	0.979***
	(0.122)	(0.101)	(0.144)
Observations	549	810	413
R^2	0.187	0.288	0.206
Log Likelihood	540.162	1,142.739	615.961

Note: *p<0.1; **p<0.05; ***p<0.01

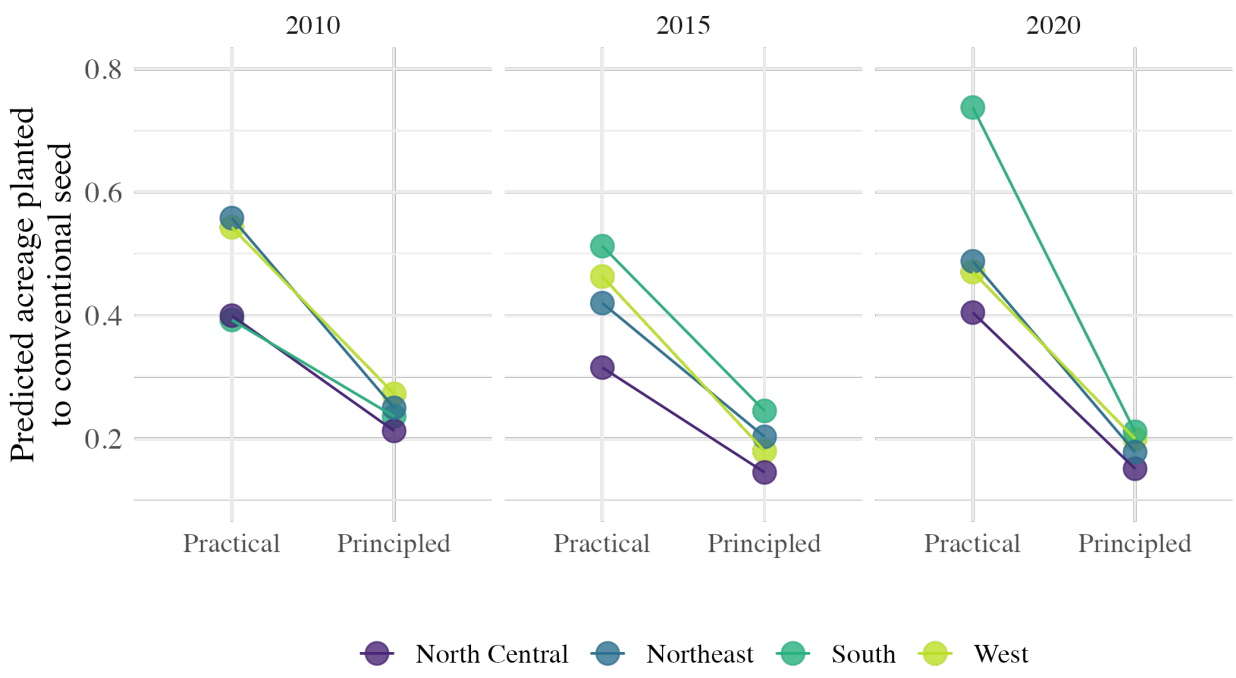


Figure 1: Predicted conventional seed use for field crop growers of two farmer profiles.

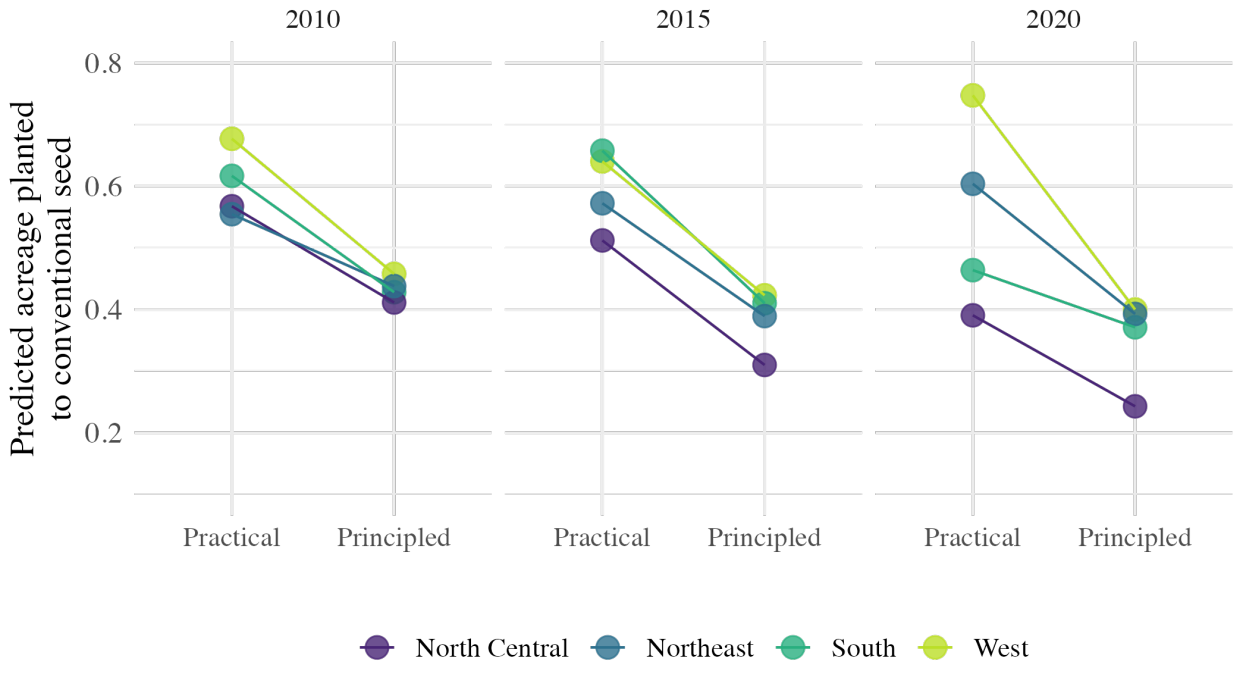


Figure 2: Predicted conventional seed use for forage crop growers of two farmer profiles.